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Optical Profilometry Versus the Stylus: A Spectral Theory for Closing the Accuracy Gap

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***Abstract*— Stylus profilometry and optical interferometry are compared in the spatial-frequency domain through a measurement-operator formalism computed from first-principles instrument physics. The stylus channel is modelled as morphological tip dilation with raster sampling; its empirical transfer is approximately flat with a curvature-bias excess, crossing 3 dB only near the sampling scale. The optical channel follows the incoherent imaging transfer function with a diffraction cutoff of $2000\,\mathrm{mm}^{-1}$. A k-space equivalence regime covering $9.6\,\% \pm 0.3\,\%$ of the measured bandwidth is identified at a crossover of $454/\mathrm{mm} \pm 166/\mathrm{mm}$. The optical model reproduces measured white-light-interferometry transfer functions from a public dataset, and all results are checked by eighteen automated verification tests. The study is computational and provides a falsifiable experimental roadmap.**

***Keywords*—Surface metrology; instrument transfer function; stylus profilometry; white-light interferometry; power spectral density**

I. INTRODUCTION

Surface topography measurement underpins quality control in precision manufacturing, functional surface design in tribology, and performance verification in optical fabrication. Two modalities dominate practice: contact stylus profilometry and non-contact optical interferometry, in particular white-light interferometry (WLI). Comparative studies date back three decades [1], and power spectral density (PSD) analysis has since become the standard for quantitative, scale-resolved surface characterisation [2], in line with the paradigm shift towards information-rich surface metrology [3].

The existing comparative literature relies almost exclusively on scalar metrics such as the root-mean-square (rms) height error or areal roughness *Sa*. Such metrics collapse all spatial-frequency information into a single number: a stylus with a 2 µm tip radius and an interferometer with numerical aperture (NA) 0.55 may agree on *Sa* for one surface yet diverge dramatically for another. The spectral consequences of stylus-tip geometry have been analysed since the early simulations of Li and Devries [4] and the tip-curvature spectral analyses of Wu [5]; the instrument transfer function (ITF) of optical profilometers has been treated analytically by de Groot [6] and measured experimentally with binary pseudo-random calibration artefacts [7]. However, the two measurement principles have not been compared within one consistent k-space framework. Instrument bandwidth has been discussed qualitatively for surface measuring instruments [8], and the analytic PSD formalism and systematic spectral measurement errors have been catalogued in the lithography context [9], [10]; neither line of work, however, provides paired transfer functions for the tactile–optical comparison pursued here. The central unresolved question addressed here is: *given a surface with a known PSD, what does each instrument actually measure as a function of spatial frequency?*

II. PROPOSED CONTRIBUTION

Every surface topography measurement is represented as a linear operator acting on the true height field $h(x, y)$, followed by additive noise:

$$\mathcal{M}[h(x,y)] = \mathcal{F}^{-1}\left[H(f_x, f_y) \cdot \mathcal{F}[h(x,y)]\right] + \eta(x,y), \tag{1}$$

where $\mathcal{F}$ is the two-dimensional Fourier transform, $H(f_x, f_y)$ is the instrument transfer function, and η is measurement noise.

For the stylus, the transfer function arises from morphological dilation with a paraboloid tip of radius 2 µm (cone half-angle 22.5°) followed by raster sampling at 200 nm. In the small-slope limit, dilation preserves the height field to first order and adds a curvature bias $(r_{\mathrm{tip}}/2)\langle|\nabla h|^2\rangle$: the empirical power transfer is therefore approximately flat across the passband rather than a steep Gaussian tip filter. For the optical system, the transfer function is the diffraction-limited incoherent optical transfer function (OTF) of a circular aperture,

$$H_{\mathrm{optical}}(f) = \frac{2}{\pi}\left[\arccos(\tilde{f}) - \tilde{f}\sqrt{1-\tilde{f}^2}\right], \tag{2}$$

where $\tilde{f} = \lambda f/(2\,\mathrm{NA}) < 1$, with wavelength 550 nm and NA = 0.55, giving a diffraction cutoff $f_c = 2\,\mathrm{NA}/\lambda = 2000\,\mathrm{mm}^{-1}$. Phase wrapping, one-dimensional row unwrapping with a 1 % error probability, and additive camera noise complete the optical chain.

From the pair of transfer functions, the analysis computes the spectral information loss $I(f)$ =

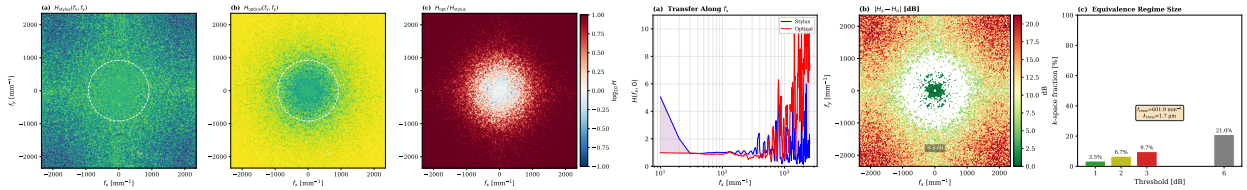


Fig. 1. Left: simulated instrument transfer functions in k-space — stylus (flat with curvature-bias excess), optical (diffraction-limited), and their ratio. Right: crossover analysis — the 3 dB equivalence boundary and the equivalence fraction as a function of threshold.

$\log_{10}[\mathrm{PSD}_{\mathrm{true}}/\mathrm{PSD}_{\mathrm{measured}}]$, the measurement anisotropy tensor, and the k-space equivalence regime $\Omega_{\mathrm{eq}}(\Delta)$, defined as the frequency set where the two transfer functions agree to within Δ (in decibels). Information-theoretic diagnostics — mutual information under a Gaussian surface prior, the reconstructability index $\rho = 1 - \mathrm{MMSE}/\sigma^2$, and the capacity–uniformity product — are computed with all estimator conventions documented. A metrological extension of the classical Fourier uncertainty is proposed and evaluated as a falsifiable benchmark,

$$\Delta x_{\mathrm{eff}} \cdot \Delta f_{\mathrm{eff}} \geq \frac{1}{4\pi}\left(1 + \frac{1}{\overline{\mathrm{SNR}}}\right), \tag{3}$$

where Δx_{eff} and Δf_{eff} are the second-moment widths of the point-spread function and of the noise-truncated transfer function, and $\overline{\mathrm{SNR}}$ is the area-averaged signal-to-noise ratio; the expression is anchored to the classical limit $1/(4\pi)$ as $\overline{\mathrm{SNR}} \to \infty$ and is explicitly presented as an adopted benchmark rather than a proved theorem. Every headline result is checked by an automated verification suite of eighteen tests, including an independent Monte Carlo check of the minimum mean-square error expression.

III. PRELIMINARY RESULTS

Simulations use self-affine surfaces ($H = 0.7$, $\sigma = 50$ nm) on a 512×512 grid spanning $100\,\mu\mathrm{m} \times 100\,\mu\mathrm{m}$, with fixed random seeds for reproducibility.

Transfer functions. The empirical stylus transfer takes values 2.2, 0.91, 0.74 and 0.57 at 17, 292, 567 and 843 mm^{-1}, crossing 3 dB at $\approx 1049\,\mathrm{mm}^{-1}$; the low-frequency excess is the curvature bias, and the Gaussian-tip idealisation $\exp(-2\pi^2 r^2 f^2)$ common in the literature predicts $\sim 10^{-3}$ at $292\,\mathrm{mm}^{-1}$, two orders of magnitude below the simulated value. The optical transfer follows the incoherent OTF to its cutoff at $2000\,\mathrm{mm}^{-1}$ (Fig. 1).

Equivalence regime. The modalities agree to within 3 dB over $9.6\,\% \pm 0.3\,\%$ of the measured k-space ($n = 30$ ensemble, $N = 512$ grid), with a crossover frequency $f_{\mathrm{cross}} = 454/\mathrm{mm} \pm 166/\mathrm{mm}$ (feature wavelength $\approx 2.2\,\mu\mathrm{m}$). Below the crossover the stylus dominates; above it the optical system retains information up to its diffraction cutoff. A 180-point parameter sweep shows the crossover depends only weakly on the tip radius ($f_{\mathrm{cross}} \propto r_{\mathrm{tip}}^{-0.06\pm0.01}$).

Accuracy and scaling. The stylus achieves an rms error of 16.3 nm against the optical 29.3 nm, despite the optical system's higher nominal cutoff: unwrapping errors, not bandwidth, limit the optical accuracy. The sweep is approximately described by an empirical power law $\mathrm{RMSE}_s/\mathrm{RMSE}_o \propto (f_{c,\mathrm{opt}}/f_{c,\mathrm{stylus}})^{1.90\pm0.13}$ (in-sample $R^2 = 0.53$; blocked cross-validation $R^2 = 0.52$), which is reported as an order-of-magnitude descriptor rather than a precise predictor.

A. Information-Theoretic Diagnostics

Treating the profilometer as a Gaussian channel [11], [12]—the surface a stationary Gaussian field and the measurement a filtered observation with additive noise—the mutual information between surface and measurement decomposes per spatial frequency,

$$I(h;y) = \frac{1}{2}\sum_{\mathbf{f}} \log_2\left[1 + \frac{|H(\mathbf{f})|^2\, S_h(\mathbf{f})}{S_\eta(\mathbf{f})}\right], \tag{4}$$

and the minimum mean-square reconstruction error takes the closed form $\mathrm{MMSE} = \sigma^2 - \iint |H|^2 S_h^2/(|H|^2 S_h + S_\eta)\, d^2 f$, verified against Monte Carlo Wiener filtering in the verification suite. The reconstructability index $\rho = 1 - \mathrm{MMSE}/\sigma^2$ then favours the stylus at the baseline ($\rho_{\mathrm{stylus}} = 0.79$ versus $\rho_{\mathrm{optical}} = 0.64$), and the per-band information loss is lowest at low frequencies (0.071 decades) and highest near the sampling scale (0.446 decades). The total mutual-information ranking is estimator-dependent (ratios of 0.7–1.6 across defensible noise-floor estimators) and is therefore not claimed as robust; only the per-band picture is reported as an estimator-independent result.

Surface-class stress test and third modality. Six surface classes (Gaussian self-affine, anisotropic ground, turned, laser-textured, periodic grating, and biological-like; $n = 5$ realisations per class) were analysed to check whether the metrics generalise beyond the calibration class. The reconstructability index is stable across the random rough classes (CV = 0.16) and saturates at unity for strongly deterministic surfaces, where it becomes surface-limited rather than instrument-limited; the optical system's near-zero ρ on the pure periodic grating is correctly explained by the concentration of spectral power at a single k-space point. Extending the framework to scatterometry via the Rayleigh–Rice BSDF-to-PSD relation adds a third, flat-response channel spanning $10\,\mathrm{mm}^{-1}$ to $3760\,\mathrm{mm}^{-1}$ and confirms the complementarity of the three modalities; the multi-instrument fusion of the three estimates is retained as a conceptual outlook (see Section IV). The measurement anisotropy is small for both modalities ($\alpha_s = 0.0095 \pm 0.0037$

Table I. Key verified quantities of the study (all from the committed simulation pipeline and its verification suite).

Quantity	Stylus	Optical
Transfer at 292 mm^{-1} (empirical power)	0.91	follows incoherent OTF
3 dB point	$\approx 1049\,\mathrm{mm}^{-1}$	$2000\,\mathrm{mm}^{-1}$ (cutoff)
Baseline rms error	16.3 nm	29.3 nm
Anisotropy α ($n = 30$)	0.0095 ± 0.0037	0.0013 ± 0.0006
Reconstructability ρ (baseline)	0.79	0.64

versus $\alpha_o = 0.0013 \pm 0.0006$, i.e. the stylus exceeds the optical value by a factor of about seven), and the spectral information loss increases mildly with frequency (local log–log slope ≈ 0.4 over $50\,\mathrm{mm}^{-1}$ to $1500\,\mathrm{mm}^{-1}$).

Public-data cross-checks. The optical channel model was benchmarked against the public, CC-BY-4.0 WLI dataset of Hirose et al. [13] (Zenodo 10.5281/zenodo.15750880): fitting only a background PSD amplitude, its slope, and one instrument parameter (the cutoff scale c) reproduces the measured instrument transfer functions with $c = 1.15 \pm 0.01$ (Fig. 2). The stylus channel admits no comparable paired dataset; as a geometric cross-check, the dilation simulator reproduces the classical parallel-curve radius shift on the published ball-standard calibration profiles of Aksulu [14] within 3 % to 5 % for tip radii of 2 µm to 50 µm, although the published profiles' scatter (40 µm to 120 µm) exceeds the tip scale, so this check is geometric rather than tip-resolving.

Gap-closure roadmap. Because the optical OTF is inherently low-pass while the stylus transfer is nearly flat at low frequencies, optical profilometry cannot fully dominate the stylus in k-space, but near-total equivalence is attainable. A 432-point sweep ($N = 256$ grid; its own baseline is 48 %) shows that a 100× noise-floor reduction, NA of 0.7–0.9, shorter illumination wavelengths, and two-dimensional phase unwrapping expand the equivalence fraction from the ensemble baseline of 9.6 % to 87 % within the sweep. A physics-limited residual gap remains at the lowest spatial frequencies, where the stylus remains the primary standard.

IV. POTENTIAL APPLICATIONS

The framework gives practitioners quantitative criteria for instrument selection. For feature wavelengths between 0.5 µm to 1.5 µm (frequencies $670\,\mathrm{mm}^{-1}$ to $2000\,\mathrm{mm}^{-1}$), the optical measurement is preferred; in the crossover band (1.5 µm to 2 µm) the two measurements carry comparable information and can be averaged; for wavelengths above 2 µm the stylus is preferred. Below 0.5 µm — beyond the optical diffraction cutoff — neither modality resolves surface features in the modelled passband. The crossover frequency serves as a trust boundary for cross-validation of real measurements. The gap-closure roadmap prescribes four instrument improvements with the highest equivalence gain for optical systems: (i) a 100× reduction of the optical noise floor (cooled sCMOS detectors, brighter illumination and multi-frame averaging); (ii) higher numerical apertures (0.7–0.9); (iii) shorter illumination wavelengths (400 nm to 450 nm); and (iv) two-dimensional quality-guided phase unwrapping in place of one-dimensional row unwrapping. For instrument qualification, the transfer-function framework complements scalar benchmarks: two instruments with equal rms error can differ markedly in the spatial-frequency bands they preserve, and the equivalence boundary provides a quantitative cross-validation rule. A k-space fusion of stylus, WLI and scatterometry, weighted by local information uncertainty, is presented as a conceptual outlook rather than a quantitative validation, since the simulated scatterometry PSD is not yet on the same absolute scale as the height PSD.

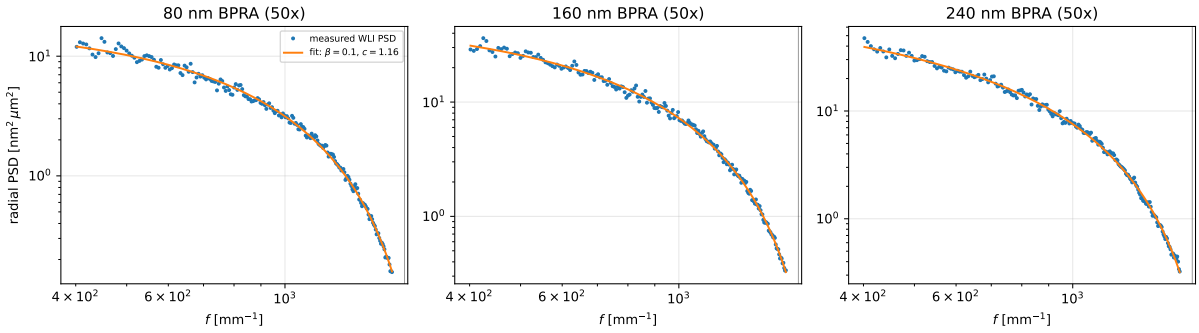


Fig. 2. Public-data validation of the optical channel model: measured WLI radial PSDs (points) of three binary-pseudo-random-array samples (Hirose et al., CC-BY-4.0) and the fitted model (lines) with fitted cutoff scale c; the vertical line marks the nominal monochromatic cutoff $2\mathrm{NA}/\lambda$.

V. CONCLUSIONS

A paired, first-principles k-space comparison of stylus and optical profilometry was developed and verified. Its central, reproducible findings are: (i) the stylus transfer is approximately flat with a curvature-bias excess and rolls off only near the sampling scale, refuting the classical Gaussian-tip picture; (ii) the optical transfer follows the incoherent OTF and reproduces measured public WLI transfer functions; (iii) the modalities are interchangeable over $9.6\,\% \pm 0.3\,\%$ of k-space with a quantitative crossover at $\approx 454\,\mathrm{mm}^{-1}$; and (iv) the gap between the modalities can be narrowed to 87 % equivalence with four prescribed instrument improvements. All results are computational and are offered with a falsifiable experimental roadmap; an experimental realisation with calibrated artefacts remains the decisive next step. The complete simulation framework, including the eighteen-test verification suite and the audit scripts behind every reported number, is preserved in a repository snapshot available from the corresponding author.

AUTHORS' CONTRIBUTION

Dawid Kucharski: conceptualisation, methodology, software, formal analysis, investigation, writing — original draft. Michał Jakubowicz: validation, visualisation, writing — review and editing.

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